

Nano-Materials for Sustainable Energy (ChE 494)

95/100

Quiz 3: Principles of Solar Energy Harvesting

April 23, 2020

40 Minutes

Total: 100

1. Sunlight is a form of "electromagnetic radiation" and the visible light that we see is a small subset of the electromagnetic spectrum.

✓



(A) True

(B) False

2. Light is a type of quantum-mechanical particle called a/an

✓

(A) Electron



(B) Photon

(C) Phonon

(D) Exciton

3. What is the relationship between the energy of a photon (E) and the wavelength of the light (λ)?

✓

$$E = hc/\lambda$$

c - Speed of light (2.998×10^8 m/s)*h* - Planck's constant (6.626×10^{-34} J.s)

4. What is photon flux?

Photon Flux is defined as the number of photons per second per unit area

✓

$$\text{Photon Flux} = (\text{Number of Photons}) / \text{m}^2.\text{s}$$

5. The Air Mass is the path length which light takes through the atmosphere normalized to the shortest possible path length (that is, when the sun is directly overhead).

✓



(A) True

(B) False

6. Semiconductors act as conductors at low temperatures and insulators at higher temperatures

✓

(A) True

☒ False

7. The band gap of a semiconductor is the minimum energy required to excite an electron that is stuck in its bound state into a free state where it can participate in conduction.

✓

☒ True

(B) False

8. Doping creates N-type material when semiconductor materials from group IV are doped withatoms.

✓

(A) Group III

(B) Group IV

☒ Group V

(D) Group VI

9. What is law of mass action?

Law of mass action states that the rate of any chemical reaction is directly proportional to the product of the active masses of the reacting substances raised to the power of their coefficients occurring in the chemical reaction.

✗

• For a Non-Reversible Reaction: $aA + bB \rightarrow \text{Products}$

Rate of Reaction (R) = $K_1 [A]^a [B]^b$

K_1 - Reaction Rate Constant

• For a Reversible Reaction: $aA + bB \rightleftharpoons cC + dD$

Rate of Forward Reaction (R_f) = $K_f [A]^a [B]^b$

K_f - Forward Reaction Rate Constant

Rate of Backward Reaction (R_b) = $K_b [C]^c [D]^d$

K_b - Backward Reaction Rate Constant

10. When the energy of a photon is equal to or greater than the band gap of the material, the photon is absorbed by the material and excites an electron into the conduction band.

✓

☒ True

(B) False

11. What is radiative (band-to-band) recombination?

✓

Radiative (band-to-band) recombination is a recombination mechanism that is dominant in direct bandgap semiconductors. An electron from the conduction band directly combines with a hole in the valence band to release a photon. The emitted photon has an energy similar to the band gap leading to its weak absorption, so that it can exit the piece of semiconductor. Example - Light Emitting Diode (LED).

12. Auger Recombination involvescarriers.

✓

(A) one



three

(C) two

(D) four

13. Auger recombination is most important at high carrier concentrations caused by heavy doping or high-level injection under concentrated sunlight.

✓



(A) True

(B) False

14. Write the current-voltage (I-V) relationship for a P-N junction diode.

For Ideal Diodes: $I = I_0 (e^{qV/kT} - 1)$

For Non Ideal Diodes: $I = I_0 (e^{qV/nkT} - 1)$

✓

I - Net Current flowing through Diode

I_0 - Dark Saturation Current (Diode Leakage Current Density in the absence of light)

q - Absolute value of Electron Charge (1.6×10^{-19} C)

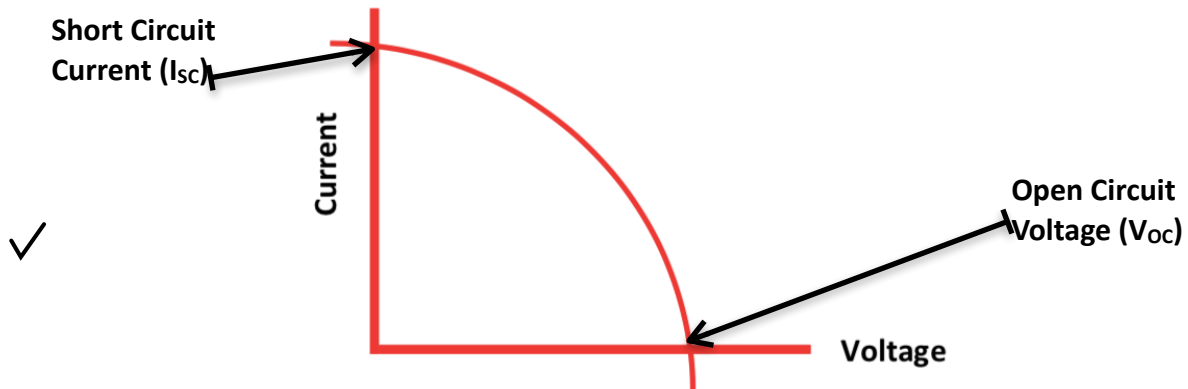
V - Applied voltage across diode terminals

k - Boltzmann's Constant (1.381×10^{-23} J/K)

T - Absolute Temperature (K)

n - Identity Factor for non-ideal diodes (inversely proportional to current) (ranges between 1-2)

15. The I-V curve of a solar cell is shown below. Indicate the short-circuit current density and open-circuit voltage?



16. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. Write the efficiency of a solar cell in terms of short-circuit current density, open-circuit voltage, fill factor, and input solar energy?

✓

$$\eta = (V_{OC} \times I_{SC} \times FF) / P_{in}$$

η - Efficiency V_{OC} - Open Circuit Voltage I_{SC} - Short Circuit Current
 FF - Fill Factor P_{in} - Input Solar Energy

17. What is the difference between direct and indirect bandgap semiconductors?

- ✓
- In a direct band gap semiconductor, the top of the valence band and the bottom of the conduction band occur at the same value of momentum.
 - In an indirect band gap semiconductor, the maximum energy of the valence band occurs at a different value of momentum to the minimum in the conduction band energy.

18. Fill-out the following table for N-type and P-type silicon semiconductor?

	N-Type Silicon	P-Type Silicon
Dopant	Group 5 elements	Group 3 elements
Majority Carriers	Electrons	Holes
Minority Carriers	Holes	Electrons

19. What is a recombination process and how many types of recombination process are there?

Recombination is a process in which an electron combines with a hole to give energy, in the form of either heat or light. Any electron which exists in the conduction band is in a meta-stable state and eventually stabilises to a lower energy position in the valence band. When this occurs, it must move into an empty valence band state. When the electron stabilises back down into the valence band, it also effectively removes a hole. This process is called recombination. It is the opposite of generation. There are 3 types of recombination - Radiative Recombination, Auger Recombination and Shockley-Read-Hall Recombination.

20. The absorption coefficient determines how far into a material light of a particular wavelength can penetrate before it is absorbed?

☒ (A) True

(B) False